

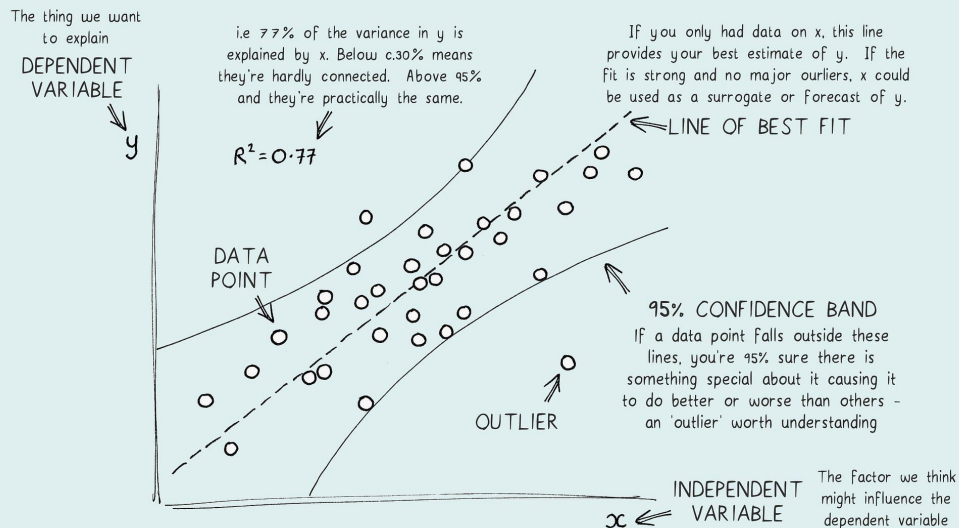
Supervised Learning 1: Regression Models

Instructor: Caryn Geady, PhD

Date: 26th January 2026

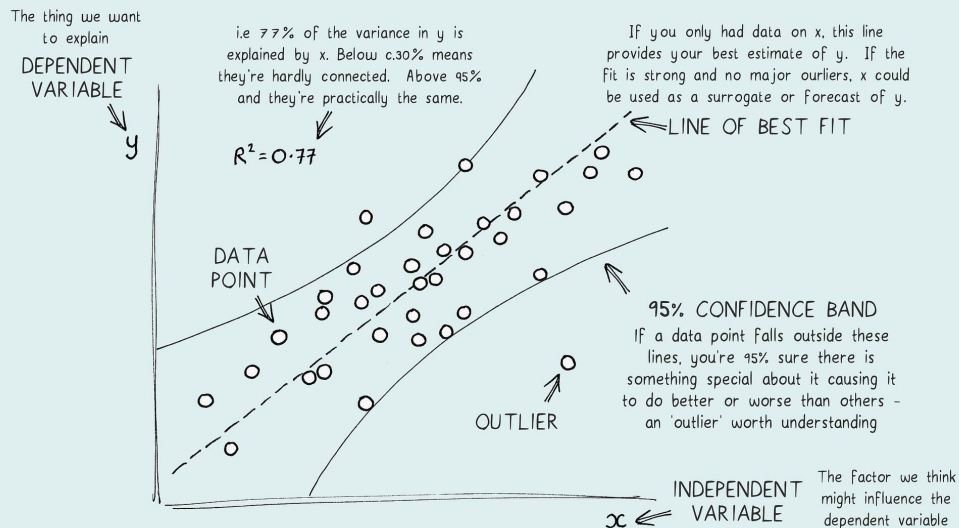
What is Regression?

- Method to **model relationships** between input and output variables.
- Types of regression:
 - Predictive modeling
 - Inferential Statistics



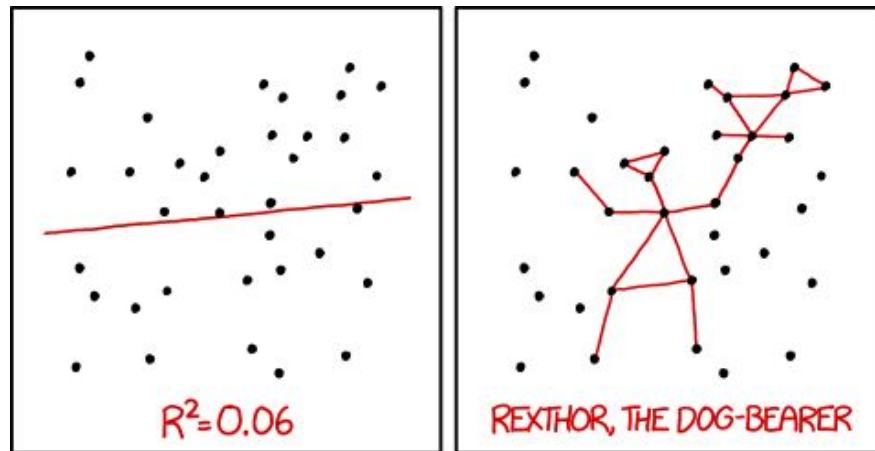
What is Regression?

- Method to **model relationships** between input and output variables.
- Types of regression:
 - Predictive modeling
 - Inferential Statistics
- Examples:
 - **Patient response to treatment** (dependent variable) as determined by **genetic markers** (independent variable)



Today's Objectives

- Introduction to Regression Models:
 - Linear;
 - Logistic;
 - Polynomial;
- Use Cases;
- Metrics for Evaluation:
 - Key Metrics: MSE, R-Squared, AUC;
- Regularization:
 - Key Techniques (L1, L2, Elastic Net)



I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

Introduction to Regression Models

Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \epsilon$$

General Equation

Linear Regression

The diagram illustrates the general equation for linear regression: $y = \beta_0 + \beta_1 x_1 + \epsilon$. Each term is color-coded and labeled with an arrow:

- dependent variable** (purple text) points to y .
- y-intercept (coefficient)** (blue text) points to β_0 .
- slope (bias)** (green text) points to β_1 .
- independent variable** (orange text) points to x_1 .
- error term (residual)** (red text) points to ϵ .

General Equation

Linear Regression

The diagram illustrates the components of a linear regression model. On the left, the **General Equation** is shown as $y = \beta_0 + \beta_1 x_1 + \epsilon$. On the right, the **Error Term** is shown as $\epsilon = y - (\beta_0 + \beta_1 x_1)$. A double-headed arrow connects the two equations. Various labels with arrows point to specific parts of the equations: 'dependent variable' points to y ; 'y-intercept (coefficient)' points to β_0 ; 'slope (bias)' points to β_1 ; 'independent variable' points to x_1 ; 'error term (residual)' points to ϵ in the general equation; 'actual value' points to y in the error term equation; and 'predicted value' points to $(\beta_0 + \beta_1 x_1)$ in the error term equation.

General Equation

$$y = \beta_0 + \beta_1 x_1 + \epsilon$$

Error Term

$$\epsilon = y - (\beta_0 + \beta_1 x_1)$$

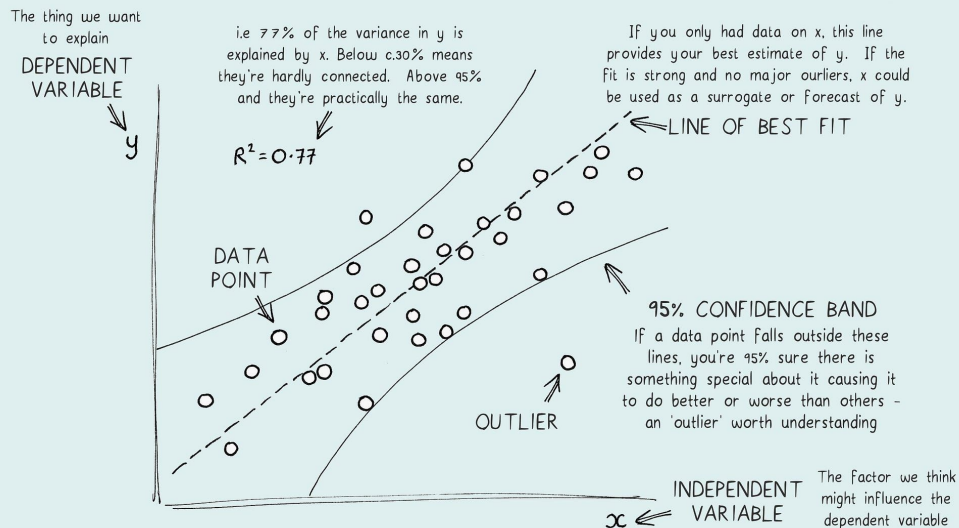
Labels and their corresponding parts:

- dependent variable: y
- y-intercept (coefficient): β_0
- slope (bias): β_1
- independent variable: x_1
- error term (residual): ϵ
- actual value: y
- predicted value: $(\beta_0 + \beta_1 x_1)$

Linear Regression



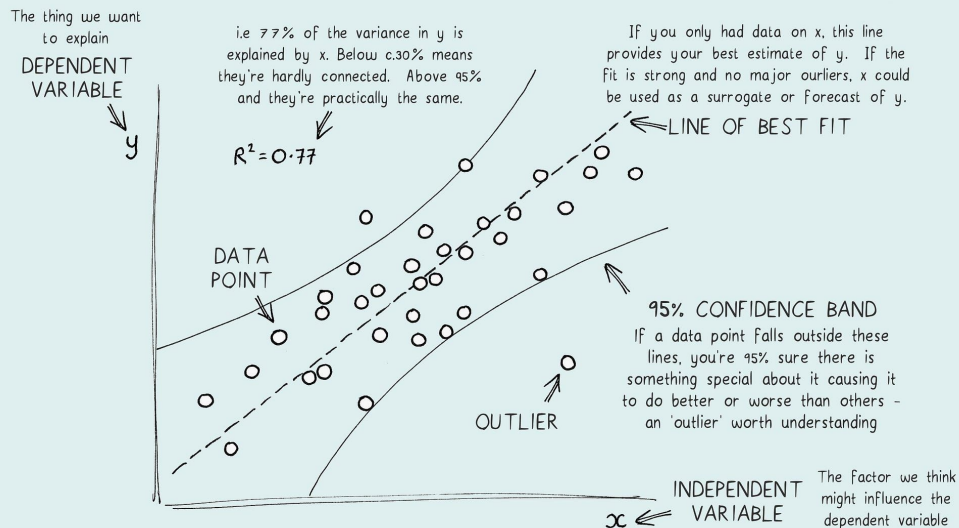
- **Key assumptions:**
 - **Linearity** - assumes a straight-line relationship between independent variables and dependent variable;
 - Independence;
 - Homoscedasticity;
 - Normality



Linear Regression

- **Key assumptions:**

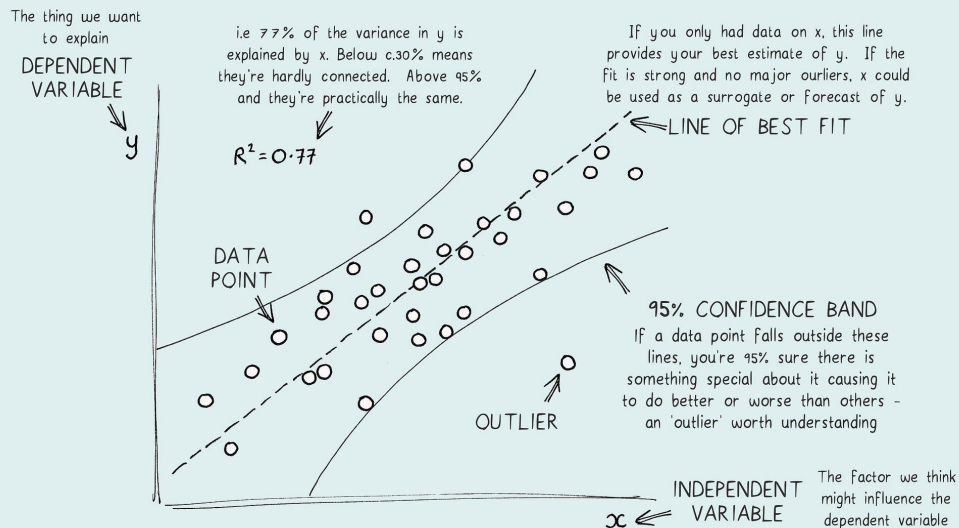
- Linearity;
- **Independence** - observations in the dataset should not influence each other;
- Homoscedasticity;
- Normality



Linear Regression

- **Key assumptions:**

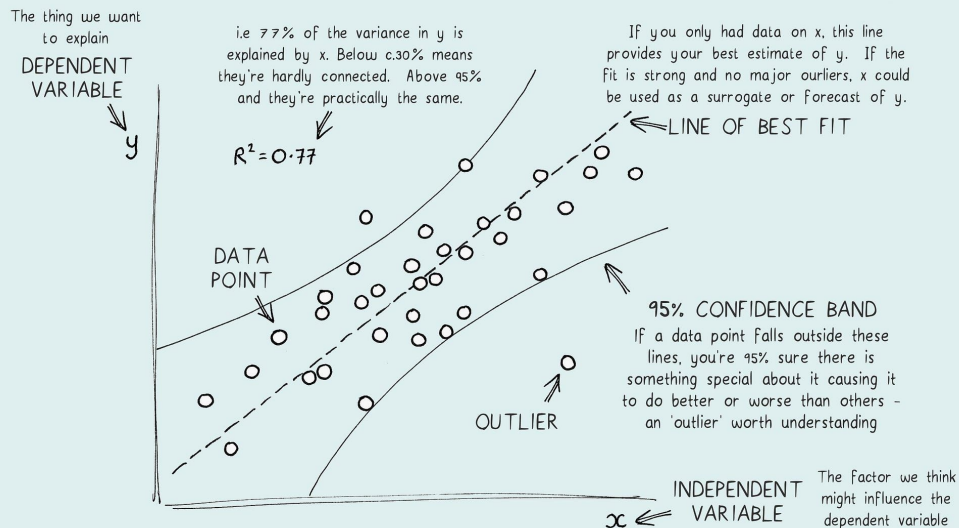
- Linearity;
- Independence;
- **Homoscedasticity** - Variance of errors (residuals) should remain constant across all levels of independent variables;
- Normality



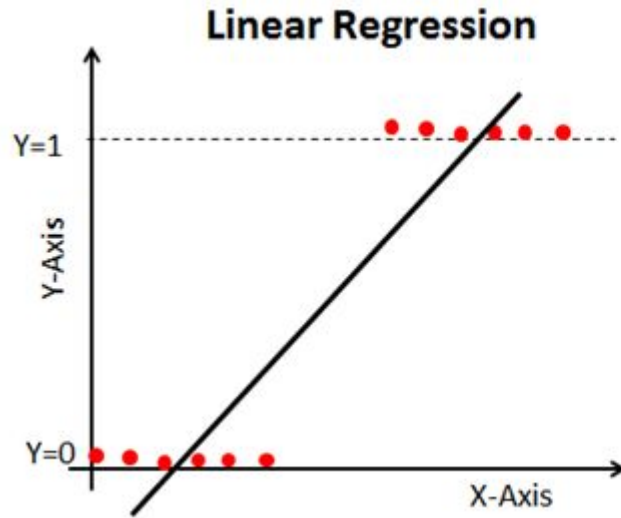
Linear Regression

- **Key assumptions:**

- Linearity;
- Independence;
- Homoscedasticity;
- **Normality** - Assumes that the errors (residuals) of the model are normally distributed

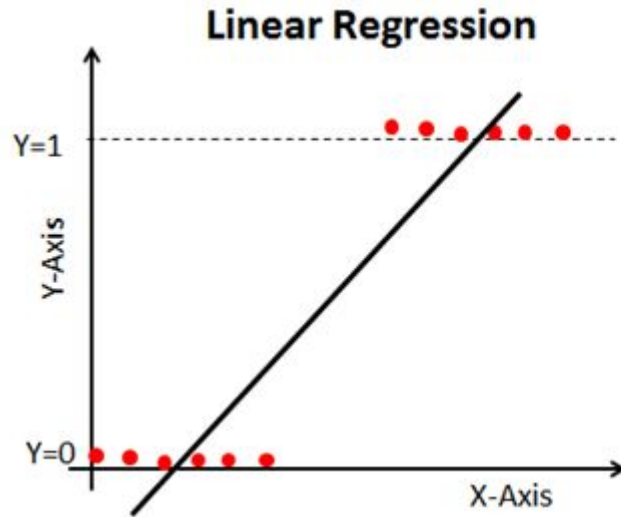


Linear Regression → Logistic Regression

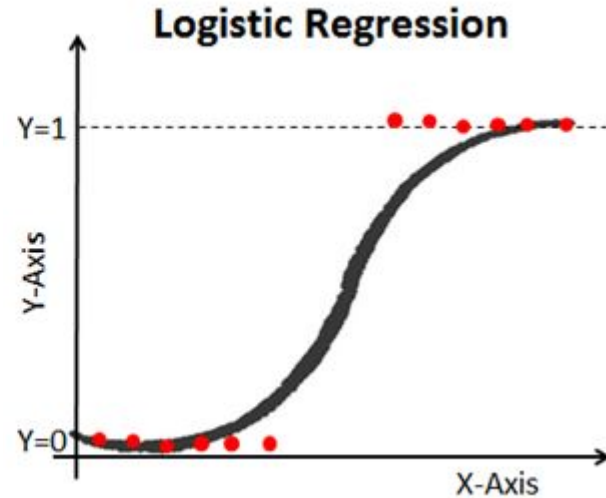


$$y = \beta_0 + \beta_1 x + \epsilon$$

Linear Regression → Logistic Regression



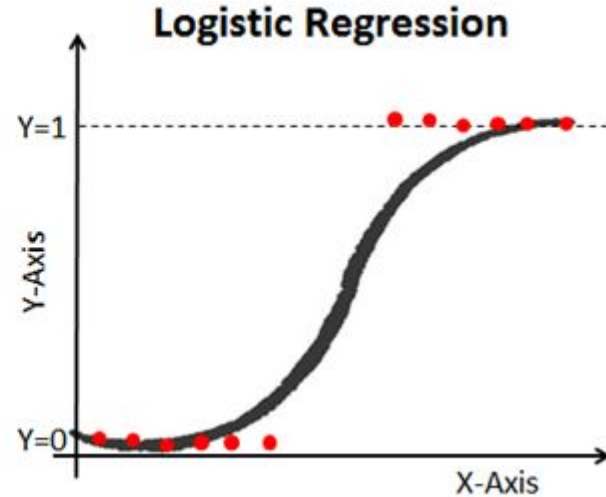
$$y = \beta_0 + \beta_1 x + \epsilon$$



$$p = \frac{1}{1 + e^{-y}}$$

Logistic Regression

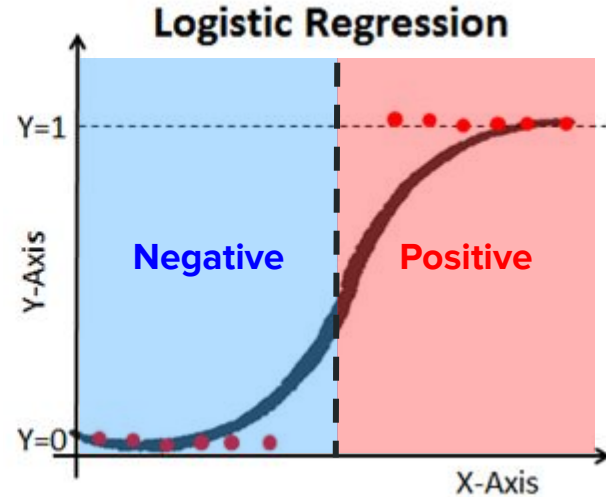
- Outputs probabilities (p) modeled with the sigmoid function;



$$p = \frac{1}{1 + e^{-y}}$$

Logistic Regression

- Outputs probabilities (p) modeled with the sigmoid function;
- Used for **classification**:
 - Binary;



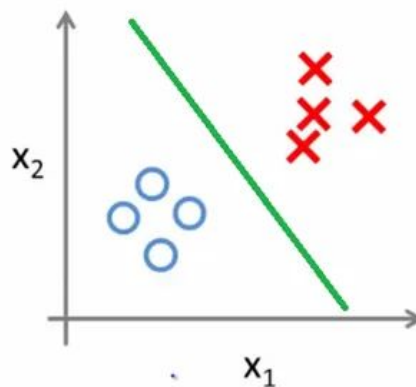
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Logistic Regression

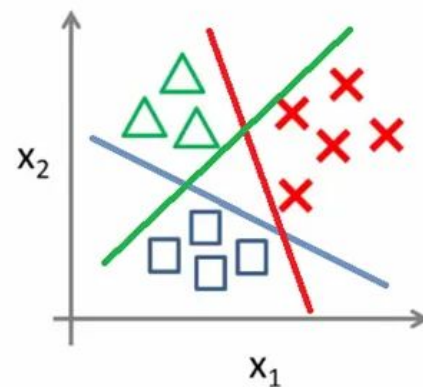


- Outputs probabilities (p) modeled with the sigmoid function;
- Used for **classification**:
 - Binary;
 - Multi-class;
- Key Differences from Linear Regression:
 - Output variable;
 - Target variable Distribution;
 - Evaluation metrics.

Binary classification:



Multi-class classification:



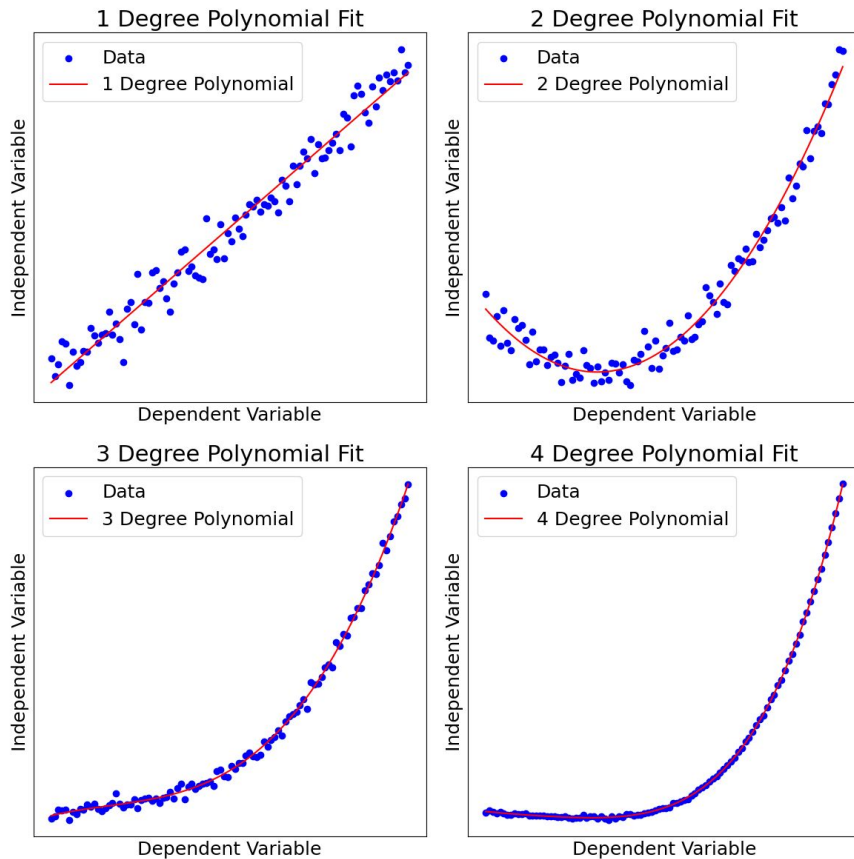


Polynomial Regression

- Extends linear regression for higher degree terms:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots$$

- Higher degree may overfit data points.



Use Cases



Regression Use Cases

- **Linear:**
 - House prices
 - Stock prices
 - Health metrics (BP,BMI)
- **Logistic:**
 - Fraud detection
 - Customer churn prediction
- **Polynomial:**
 - Disease progression
 - Health metrics (circadian rhythm)

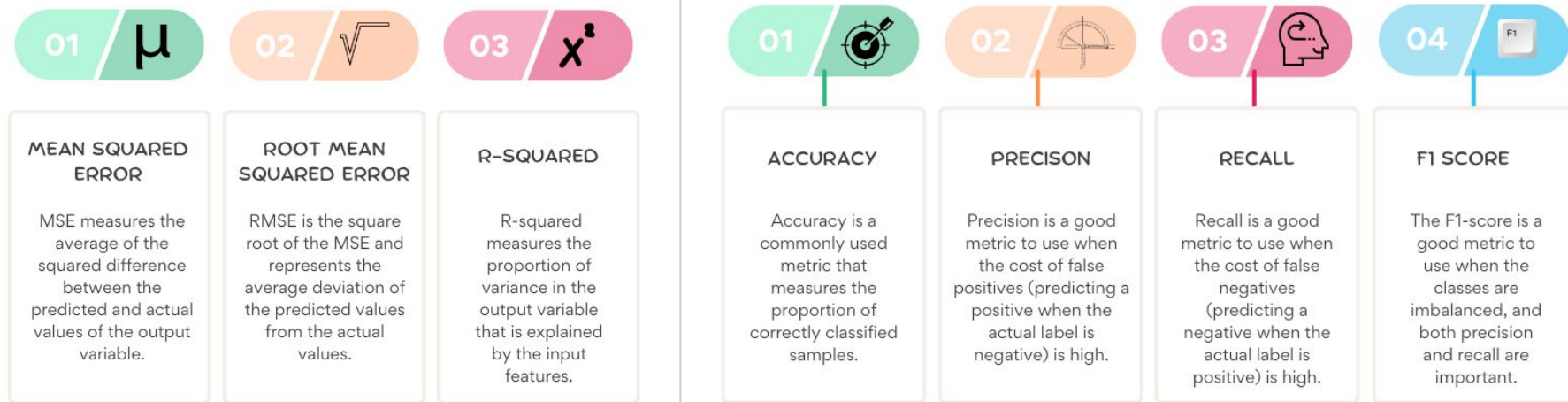


Metrics for Evaluation



Evaluating Regression Models

- Objective evaluation of model performance → metric selection is crucial!
- Some examples:



Regression Classification

Evaluating Classification Models

- Objective evaluation of model performance → metric selection is crucial!
- Some examples:



MEAN SQUARED ERROR

MSE measures the average of the squared difference between the predicted and actual values of the output variable.



ROOT MEAN SQUARED ERROR

RMSE is the square root of the MSE and represents the average deviation of the predicted values from the actual values.



R-SQUARED

R-squared measures the proportion of variance in the output variable that is explained by the input features.

Regression



ACCURACY

Accuracy is a commonly used metric that measures the proportion of correctly classified samples.



PRECISION

Precision is a good metric to use when the cost of false positives (predicting a positive when the actual label is negative) is high.



RECALL

Recall is a good metric to use when the cost of false negatives (predicting a negative when the actual label is positive) is high.



F1 SCORE

The F1-score is a good metric to use when the classes are imbalanced, and both precision and recall are important.

Classification

NEXT WEEK

Evaluation Metrics for Regression

- Recall the error term (for a single observation) $\rightarrow \varepsilon = y - (\beta_0 + \beta_1 x_1)$
- Let the predicted value $\rightarrow \hat{y} = \beta_0 + \beta_1 x_1$

01

μ

MEAN SQUARED ERROR

MSE measures the average of the squared difference between the predicted and actual values of the output variable.

02

$\sqrt{\quad}$

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x^2

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R-squared measures the proportion of variance in the output variable that is explained by the input features.

Evaluation Metrics for Regression

- Recall the loss function (for a single observation) $\rightarrow \varepsilon = y - (\beta_0 + \beta_1 x_1)$
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R-squared measures the proportion of variance in the output variable that is explained by the input features.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

Evaluation Metrics for Regression

- Recall the loss function (for a single observation) $\rightarrow \varepsilon = y - (\beta_0 + \beta_1 x_1)$
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R-squared measures the proportion of variance in the output variable that is explained by the input features.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2}$$

Evaluation Metrics for Regression

- Recall the loss function (for a single observation) $\rightarrow \varepsilon = y - (\beta_0 + \beta_1 x_1)$
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R-SQUARED

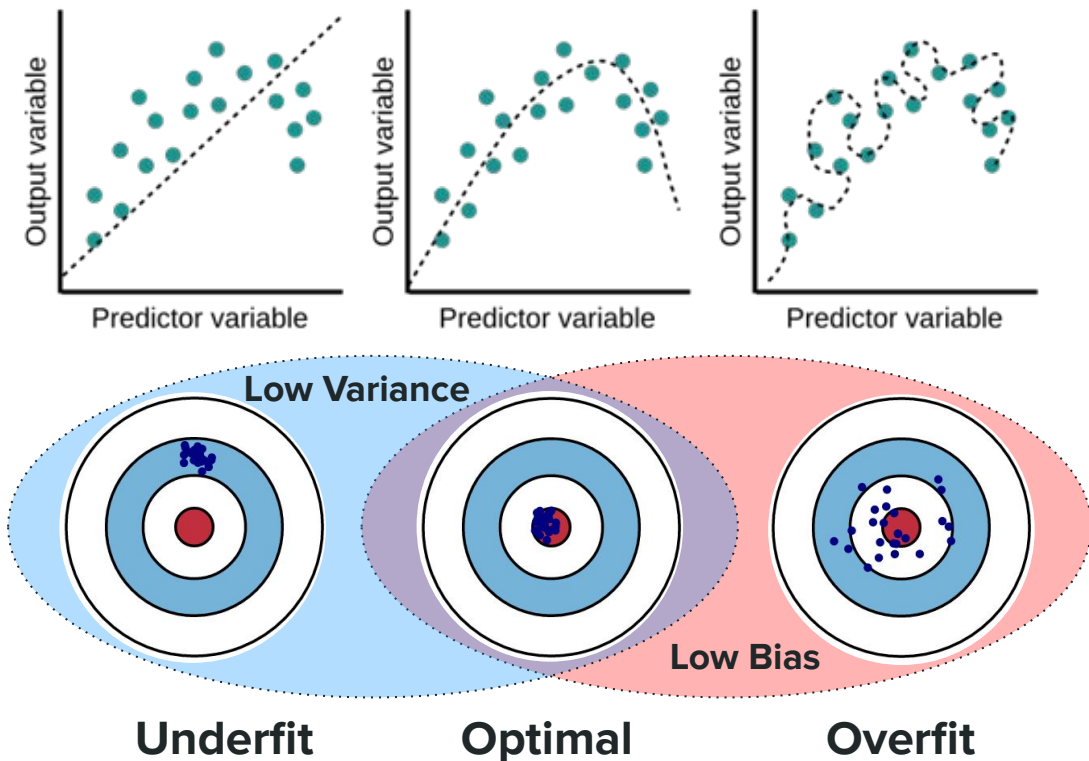
R-squared measures the proportion of variance in the output variable that is explained by the input features.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y})^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Regularization

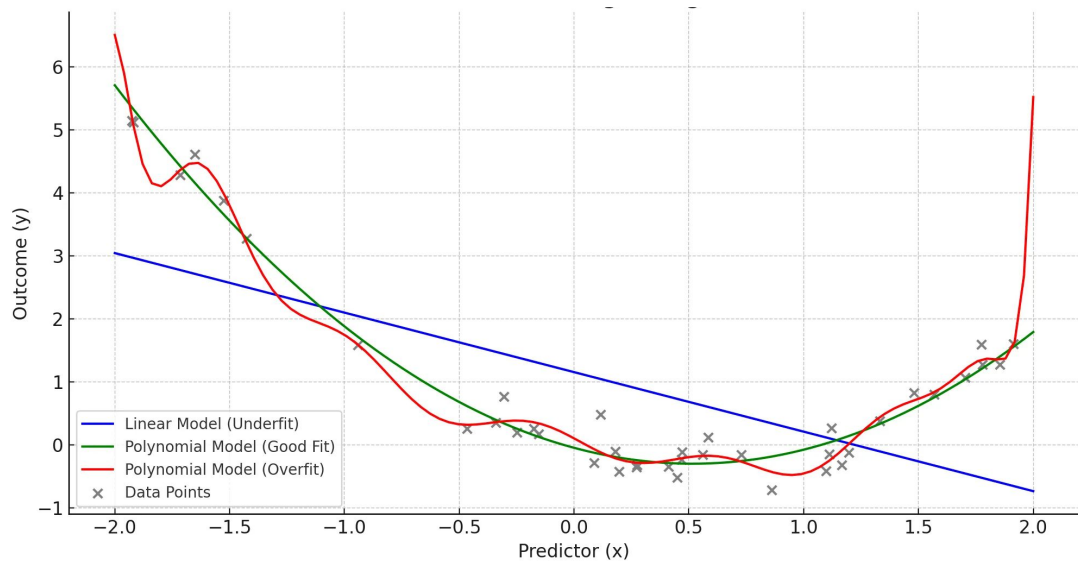
The Importance of Splitting Your Data (RECAP)

- Data splitting ensure models are evaluated on data they have not seen before;
- Goal → strike a balance between complexity and simplicity:
 - Overfitting (too complex) memorizes training data but fails to generalize;
 - Underfitting (too simple) fails to capture critical patterns in the data



What is Regularization?

- Prevents overfitting by adding penalties to large coefficients; 
- Standard $\rightarrow$ minimize the error (residual)
- Regularization $\rightarrow$ minimize the error **PLUS** penalty term



Key Techniques



L1 Regularization (Lasso)

- Adds penalty proportional to the **absolute value** of the coefficients
- Drives some coefficients to zero, enabling **feature selection**

L2 Regularization (Ridge)

- Adds penalty proportional to the **square** of the coefficients;
- Shrinks coefficients toward zero, but does not eliminate them entirely

Elastic Net

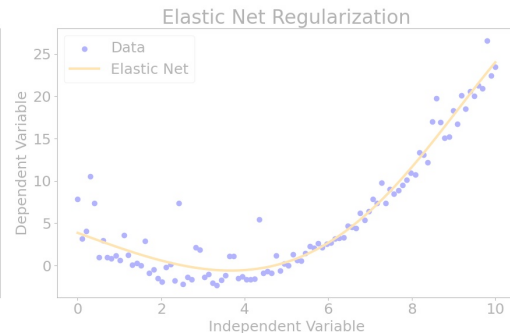
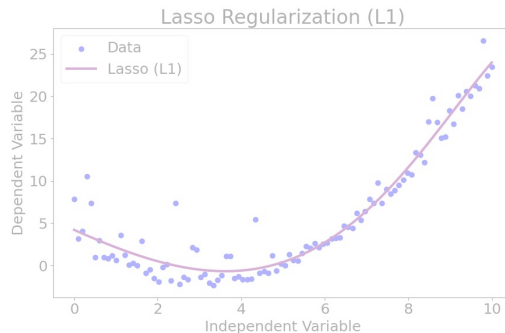
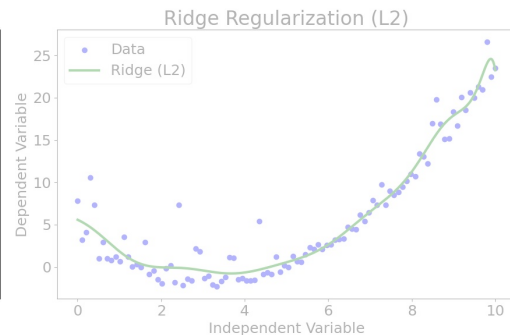
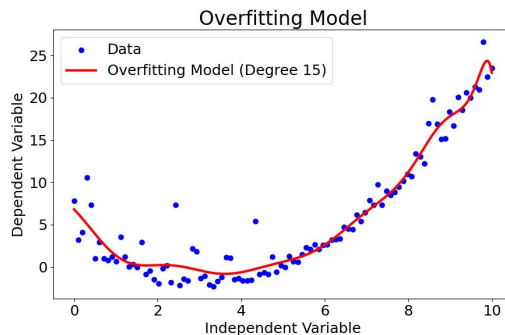
- Combines L1 and L2 penalties to **balance** feature selection and coefficient shrinkage;
- Common for datasets with correlated features

Comparing Techniques

- **Overfitting Model**

(Top-Left):

- Demonstrates overfitting with a high-degree polynomial (Degree 15).
- The model captures noise in the data rather than the underlying pattern.
- Result: Poor generalization to unseen data.

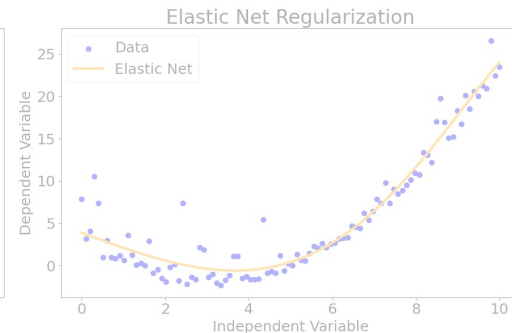
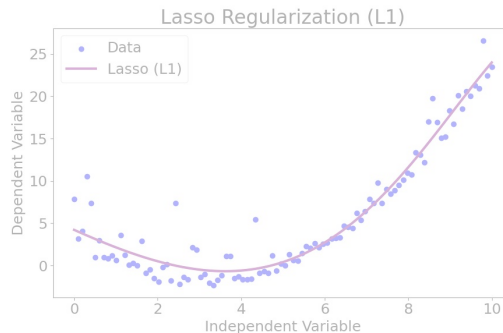
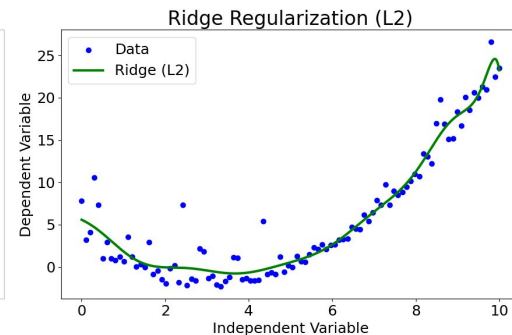
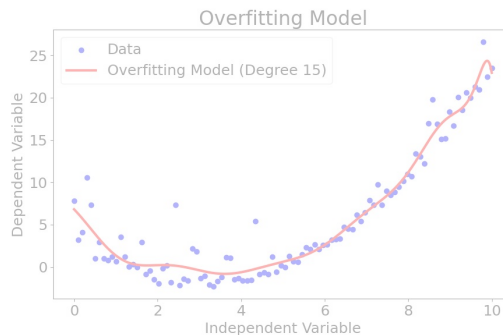


Comparing Techniques

- **Ridge Regularization**

- (Top-Right):**

- Applies L2 regularization to shrink coefficients.
- Reduces overfitting while retaining key trends in the data.
- Best for cases with correlated features.

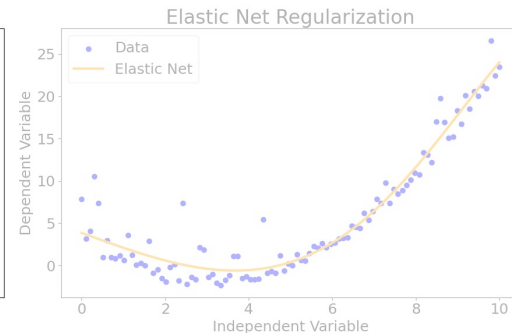
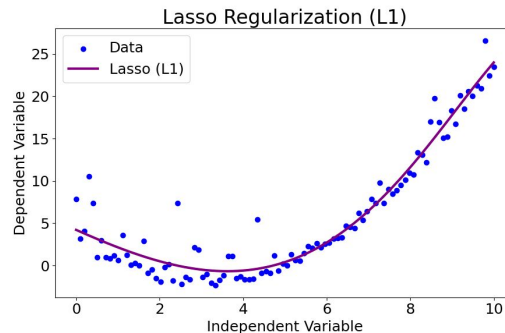
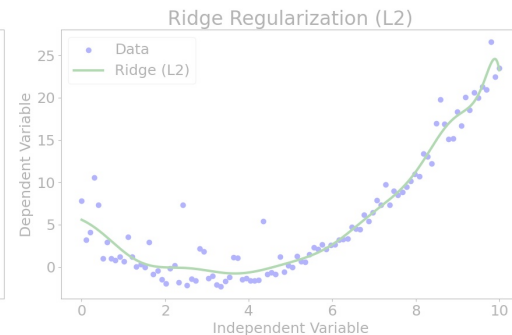
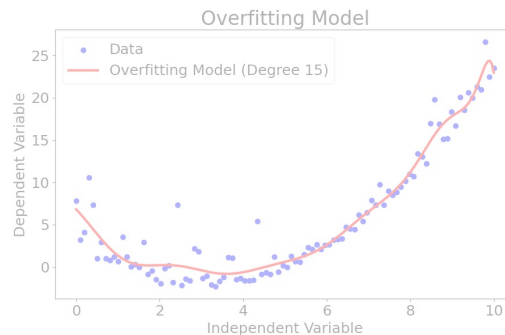


Comparing Techniques

- **Lasso Regularization**

- (Bottom-Left):**

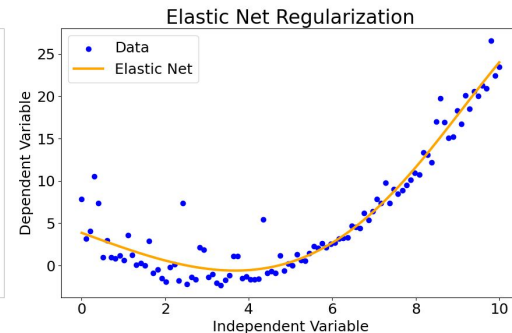
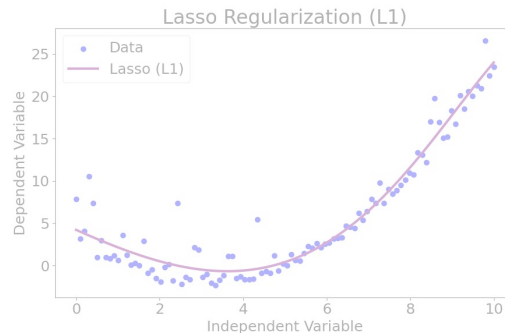
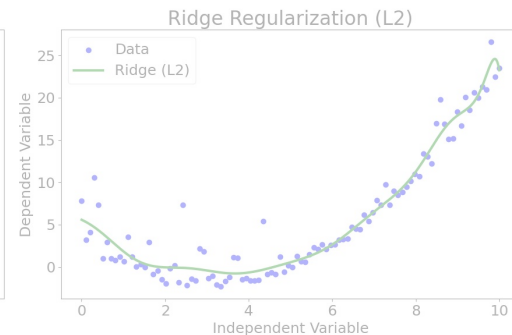
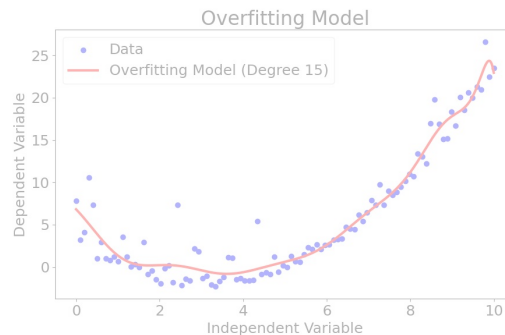
- Applies L1 regularization to shrink some coefficients to zero.
- Facilitates feature selection and simplifies the model.
- Best for sparse datasets with irrelevant features.



Comparing Techniques

- **Elastic Net Regularization (Bottom-Right):**

- Combines L1 and L2 regularization.
- Balances feature selection and coefficient shrinkage.
- Ideal for datasets with highly correlated predictors.



Summary and Q&A



Summary

Category	Type/Metric	When to Use	When Not to Use
Regression Types	Linear Regression	- Predicting continuous outcomes with a linear relationship.	- When data has non-linear patterns or interactions.
	Logistic Regression	- Predicting binary or categorical outcomes (e.g., 0/1, true/false).	- When target variable is continuous or the relationship is highly non-linear.
	Polynomial Regression	- When the relationship between variables is non-linear but can be expressed with polynomial terms.	- When data is high-dimensional, as it increases overfitting risk.

Summary

Category	Type/Metric	When to Use	When Not to Use
Evaluation Metrics	MSE (Mean Squared Error)	- Assessing model accuracy for continuous targets (lower = better).	- When evaluating classification tasks (doesn't apply).
	RMSE (Root Mean Squared Error)	- Measuring prediction error for continuous targets (lower = better).	- When evaluating classification tasks or comparing models where interpretability is more important.
	R-squared	- Measuring how well the model explains variance in continuous target variables.	- When comparing models with different numbers of predictors or in classification problems.

Summary

Category	Type/Metric	When to Use	When Not to Use
Regularization Methods	L1 (Lasso)	- Feature selection (shrinks less relevant coefficients to 0).	- When you don't want to exclude any feature, or multicollinearity is a major issue.
	L2 (Ridge)	- Preventing overfitting by shrinking coefficients, works well with correlated predictors.	- When sparse solutions are needed or feature selection is important.
	Elastic Net	- Combining L1 and L2 for feature selection with correlated predictors.	- When the simplicity of pure L1 or L2 regularization is sufficient or computational efficiency is a concern.

Questions ?

"the best way to learn a language
is to speak to natives"
the guy learning Python:

